

AI & ML Summer Internship 2026

Assignment 10 - Model Evaluation

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Objective

To evaluate a Logistic Regression model using Accuracy, Precision, Recall, F1-Score and Confusion Matrix, compare evaluation metrics graphically, and understand their significance.

Exercise 1 - Model Evaluation using Logistic Regression

Python Code:

```
In [2]: #####
# Assignment 10 - Exercise 1
# Model Evaluation using Logistic Regression
#####

import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import (
    accuracy_score,
    precision_score,
    recall_score,
    f1_score,
    confusion_matrix,
    ConfusionMatrixDisplay,
    classification_report
)

# -----
# Step 1 : Load Dataset
# -----

titanic = pd.read_csv(r"C:\Users\91953\titanic.csv")

print("\n" + "="*60)
print("Original Titanic Dataset")
print("\n" + "="*60)
print(titanic.head())

# -----
# Step 2 : Select Features and Target
# -----

X = titanic[['Age', 'Fare', 'Sex', 'Pclass']]
y = titanic['Survived']

# -----
# Step 3 : Handle Missing Values
# -----

X = X.fillna(X.mean())

print("\n" + "="*60)
print("Selected Features")
print("\n" + "="*60)
print(X.head())

print("\nTarget Variable")
print(y.head())

# -----
# Step 4 : Split Dataset
# -----
```

```

# Step 7 : Evaluation Metrics
# -----

accuracy = accuracy_score(y_test, y_pred)
precision = precision_score(y_test, y_pred)
recall = recall_score(y_test, y_pred)
f1 = f1_score(y_test, y_pred)

print("\n" + "="*60)
print("Model Evaluation")
print("="*60)

print(f"Accuracy : {accuracy:.4f}")
print(f"Precision : {precision:.4f}")
print(f"Recall : {recall:.4f}")
print(f"F1 Score : {f1:.4f}")

print("\nClassification Report")
print(classification_report(y_test, y_pred))

# -----
# Step 8 : Confusion Matrix
# -----

cm = confusion_matrix(y_test, y_pred)

print("\nConfusion Matrix")
print(cm)

# -----
# Step 9 : Plot Confusion Matrix
# -----

plt.figure(figsize=(6,6))

disp = ConfusionMatrixDisplay(
    confusion_matrix=cm,
    display_labels=["Not Survived", "Survived"]
)

disp.plot(cmap="Blues")

plt.title("Confusion Matrix - Logistic Regression")

plt.grid(False)

plt.show()

# -----
# Step 10 : Observations
# -----

print("\n" + "="*60)
print("Observations")
print("="*60)

print("1. Accuracy represents the overall prediction correctness.")
print("2. Precision indicates how many predicted survivors were actually survivors.")
print("3. Recall measures how many actual survivors were identified.")
print("4. F1-Score provides a balance between Precision and Recall.")
print("5. Confusion Matrix summarizes correct and incorrect predictions.")

print("\nExercise 1 Completed Successfully.")

```

Execution Output:

```

Original Titanic Dataset
=====
 PassengerId  Age   Fare  Sex  sibsp  zero  zero.1  zero.2  zero.3 \
0            1  22.0   7.2500  0    1    0    0    0    0
1            2  38.0  71.2833  1    1    0    0    0    0
2            3  26.0   7.9250  1    0    0    0    0    0
3            4  35.0  53.1000  1    1    0    0    0    0
4            5  35.0   8.0500  0    0    0    0    0    0

 zero.4  ...  zero.12  zero.13  zero.14  Pclass  zero.15  zero.16  Embarked \
0        0  ...        0        0        0        3        0        0        2.0
1        0  ...        0        0        0        1        0        0        0.0
2        0  ...        0        0        0        3        0        0        2.0
3        0  ...        0        0        0        1        0        0        2.0
4        0  ...        0        0        0        3        0        0        2.0

 zero.17  zero.18  Survived
0         0         0        0
1         0         0        1
2         0         0        1
3         0         0        1
4         0         0        0

[5 rows x 28 columns]

=====
Selected Features
=====
   Age   Fare  Sex  Pclass
0  22.0   7.2500  0    3
1  38.0  71.2833  1    1
2  26.0   7.9250  1    3
3  35.0  53.1000  1    1
4  35.0   8.0500  0    3

Target Variable
0    0
1    1
2    1
3    1
4    0
Name: Survived, dtype: int64

Training Samples : 1047
Testing Samples  : 262

=====
Model Evaluation
=====
Accuracy : 0.7672
Precision : 0.6579
Recall : 0.3425
F1 Score : 0.4505

```

```

Classification Report
              precision    recall  f1-score   support

     0       0.79        0.93        0.85        189
     1       0.66        0.34        0.45         73

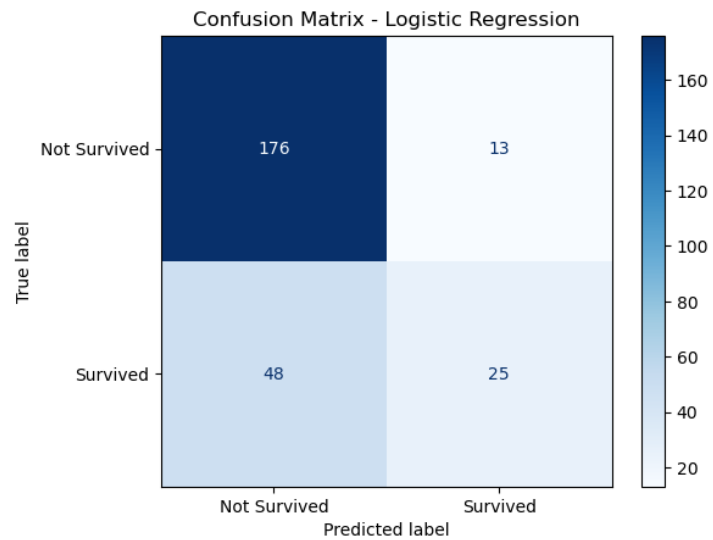
 accuracy          0.77        262
 macro avg          0.72        0.64        0.65        262
 weighted avg       0.75        0.77        0.74        262

```

```

Confusion Matrix
[[176  13]
 [ 48  25]]
<Figure size 600x600 with 0 Axes>

```



```

=====
Observations
=====
1. Accuracy represents the overall prediction correctness.
2. Precision indicates how many predicted survivors were actually survivors.
3. Recall measures how many actual survivors were identified.
4. F1-Score provides a balance between Precision and Recall.
5. Confusion Matrix summarizes correct and incorrect predictions.

Exercise 1 Completed Successfully.

```

Exercise 2 - Comparison of Evaluation Metrics

Python Code:

```
jupyter Exercise-02-Metric-Comparison.ipynb Last Checkpoint: 1 minute ago
File Edit View Run Kernel Settings Help Trusted
JupyterLab Python 3 (ipykernel)

import matplotlib.pyplot as plt

# -----
# Step 1 : Model Evaluation Scores
# -----

accuracy = 0.7672
precision = 0.6579
recall = 0.3425
f1 = 0.4585

print("="*60)
print("Evaluation Metrics")
print("="*60)

print(f"Accuracy : {accuracy:.4f}")
print(f"Precision : {precision:.4f}")
print(f"Recall : {recall:.4f}")
print(f"F1 Score : {f1:.4f}")

# -----
# Step 2 : Store Metrics
# -----

metrics = ['Accuracy', 'Precision', 'Recall', 'F1-Score']
values = [accuracy, precision, recall, f1]

# -----
# Step 3 : Plot Bar Graph
# -----

plt.figure(figsize=(8,6))

bars = plt.bar(
    metrics,
    values,
    edgecolor='black',
    linewidth=1.2
)

plt.title(
    "Comparison of Model Evaluation Metrics",
    fontsize=15,
    fontweight='bold'
)

plt.xlabel(
    "Evaluation Metrics",
    fontsize=12
)

plt.ylabel(
    "Score",
    fontsize=12
)

plt.ylim(0,1)

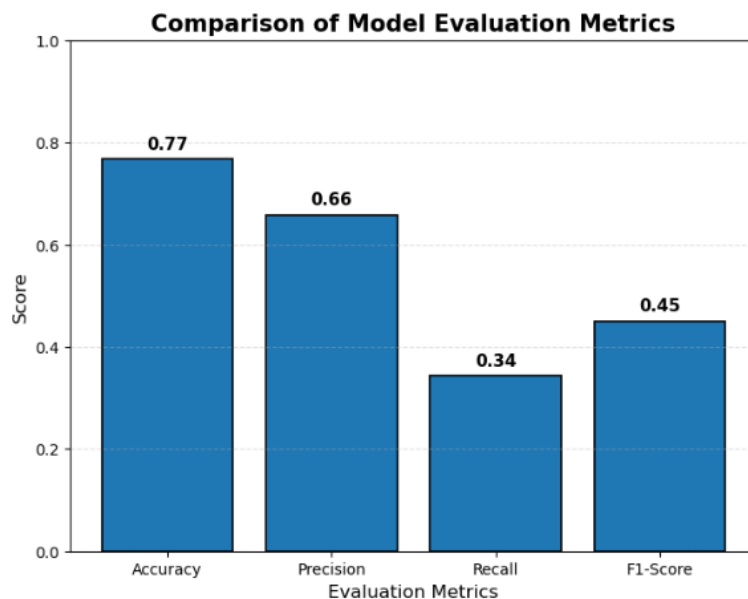
plt.grid(axis='y', linestyle='--', alpha=0.4)

# -----
# Step 4 : Display Values on Bars
# -----

for bar in bars:
    height = bar.get_height()
    plt.text(
        bar.get_x() + bar.get_width()/2,
        height + 0.02,
        f"{height:.2f}",
        ha='center',
        fontsize=11,
        fontweight='bold'
    )
```

Execution Output:

```
=====
Evaluation Metrics
=====
Accuracy : 0.7672
Precision : 0.6579
Recall : 0.3425
F1 Score : 0.4505
```



```
=====
Observations
=====
```

1. Accuracy is the highest evaluation metric, indicating good overall prediction performance.
2. Precision is reasonably high, showing that many predicted survivors were actually correct.
3. Recall is comparatively low, indicating that some actual survivors were missed by the model.
4. F1-Score provides a balanced evaluation by considering both Precision and Recall.
5. The bar graph makes it easy to compare all evaluation metrics visually.
6. Logistic Regression performs reasonably well but can be improved with better feature selection or advanced models.

Exercise 2 Completed Successfully.

Exercise 3 - Reflection

What is Accuracy?

Accuracy is the percentage of correct predictions made by a machine learning model. It provides an overall measure of performance but may be misleading for imbalanced datasets.

Explain Precision and Recall.

Precision measures how many predicted positive cases are actually positive, while Recall measures how many actual positive cases are correctly identified. Both metrics evaluate different aspects of classification.

Why is F1-Score important?

F1-Score is the harmonic mean of Precision and Recall. It provides a balanced evaluation when both false positives and false negatives are important.

Why is the Confusion Matrix useful?

The Confusion Matrix summarizes True Positives, True Negatives, False Positives and False Negatives. It helps identify model strengths and weaknesses.

Reflection

This assignment improved my understanding of model evaluation. I learned to interpret Accuracy, Precision, Recall, F1-Score and Confusion Matrix together instead of relying on a single metric.

Conclusion

Model evaluation is an essential stage of machine learning. Using multiple evaluation metrics provides a more reliable understanding of model performance and helps improve prediction quality.